GLASS-BOX GREEN INSIGHTS

# Final JSON Contract

Version 1.0.0

**One object = one customer + one green-product opportunity**

This is the shared integration contract for data, ML, eligibility, carbon calculations, recommendations, LLM explanations, dashboard and audit logging.

## Status: FINAL FOR IMPLEMENTATION

Any later structural change must increase schema_version and be agreed by all component owners.

# 1. How the team should use it

Every component reads its required fields and writes only the section it owns. The completed object moves forward through the pipeline; later components must not rewrite earlier decisions.

| Field | Allowed values |
|---|---|
| Data pipeline | customer, opportunity, data_cutoff_date |
| ML | prediction |
| Eligibility engine | eligibility |
| Carbon engine | impact |
| Recommendation engine | recommendation |
| LLM + verifier | explanation |
| Integration layer | request metadata and audit |

**The LLM may read the object, but it may write only explanation.**

## Storage rule

Store one record for every customer_id + product_code combination. A customer therefore normally has four records: EV Finance, Green Home Loan, Solar Finance and Paperless Banking.

# 2. Complete canonical example

The following is a successful EV Finance opportunity. Values are illustrative; the field names and structure are the contract.


```
{
  "schema_version": "1.0.0",
  "request_id": "REQ-20261010-000001",
  "generated_at": "2026-10-10T18:30:00+05:30",
  "data_cutoff_date": "2026-08-31",
  "customer": {
    "customer_id": "C10001",
    "segment": "SUBURBAN_HOMEOWNER",
    "region": "LONDON",
    "data_completeness": 0.92
  },
  "opportunity": {
    "product_id": "GP01",
    "product_code": "EV_FINANCE",
    "product_name": "EV Finance",
    "category": "TRANSPORT"
  },
  "prediction": {
    "propensity": 0.84,
    "confidence": 0.87,
    "confidence_band": "HIGH",
    "model_version": "ev_propensity_v1.0.0",
    "reason_codes": [
      {
        "code": "HIGH_FUEL_SPEND",
        "label": "High recurring fuel spending",
        "importance": 0.38,
        "evidence": {
          "metric": "average_monthly_fuel_spend",
          "value": 245.5,
          "unit": "GBP",
          "comparison": "ABOVE_SEGMENT_AVERAGE"
        }
      },
      {
        "code": "HIGH_ANNUAL_MILEAGE",
        "label": "High annual vehicle mileage",
        "importance": 0.31,
        "evidence": {
          "metric": "annual_mileage",
          "value": 16500,
          "unit": "miles/year",
          "comparison": "ABOVE_THRESHOLD"
        }
      },
      {
        "code": "EXISTING_ICE_VEHICLE",
        "label": "Currently owns a petrol or diesel vehicle",
        "importance": 0.19,
        "evidence": {
          "metric": "vehicle_fuel_type",
          "value": "PETROL",
          "unit": null,
          "comparison": null
        }
      }
    ]
  },
  "eligibility": {
    "status": "ELIGIBLE",
    "is_eligible": true,
    "rule_version": "ev_eligibility_v1.0.0",
    "decision_codes": [
      {
        "code": "HAS_EXISTING_VEHICLE",
        "result": "PASS",
        "message": "Customer has an existing vehicle."
      },
      {
        "code": "NOT_EXISTING_EV_OWNER",
        "result": "PASS",
        "message": "No existing electric vehicle was found."
      },
      {
        "code": "MILEAGE_DATA_AVAILABLE",
        "result": "PASS",
        "message": "Annual mileage information is available."
      }
```

```
    ],
    "failed_rules": [],
    "missing_requirements": [],
    "requires_human_review": false
  },
  "impact": {
    "status": "AVAILABLE",
    "method_id": "EV_AVOIDED_EMISSIONS_V1",
    "method_name": "Current fuel emissions minus estimated EV electricity emissions",
    "methodology_reference": "GHG_PROTOCOL_AVOIDED_EMISSIONS",
    "time_horizon": {
      "value": 1,
      "unit": "YEAR"
    },
    "primary_metric": {
      "metric": "POTENTIAL_CO2E_AVOIDED",
      "lower_estimate": 900.0,
      "central_estimate": 1050.0,
      "upper_estimate": 1200.0,
      "unit": "kgCO2e/year"
    },
    "supporting_metrics": [
      {
        "metric": "CURRENT_TRANSPORT_EMISSIONS",
        "value": 2850.0,
        "unit": "kgCO2e/year"
      },
      {
        "metric": "ESTIMATED_EV_EMISSIONS",
        "value": 1800.0,
        "unit": "kgCO2e/year"
      }
    ],
    "assumptions": [
      {
        "code": "MILEAGE_UNCHANGED",
        "description": "Annual mileage remains unchanged after switching vehicles."
      },
      {
        "code": "EV_EFFICIENCY_RANGE",
        "description": "EV energy efficiency is represented using a documented range."
      }
    ],
    "emission_factors": [
      {
        "factor_id": "UK_PETROL_2026",
        "activity": "PETROL_COMBUSTION",
        "value": 2.31,
        "unit": "kgCO2e/litre",
        "source": "UK_DESNZ_DEFRA",
        "source_year": 2026,
        "factor_version": "2026.1"
      },
      {
        "factor_id": "UK_GRID_2026",
        "activity": "GRID_ELECTRICITY",
        "value": 0.18,
        "unit": "kgCO2e/kWh",
        "source": "UK_DESNZ_DEFRA",
        "source_year": 2026,
        "factor_version": "2026.1"
      }
    ],
    "calculation_inputs": {
      "annual_mileage": 16500,
      "annual_fuel_litres": 1234.0,
      "estimated_ev_kwh": 10000.0
    },
    "calculation_timestamp": "2026-10-10T18:29:30+05:30"
  },
  "recommendation": {
    "decision": "RECOMMEND",
    "priority_score": 0.73,
    "priority_band": "HIGH",
    "action_code": "EXPLORE_EV_OPTIONS",
    "action_label": "Explore EV options",
    "customer_visible": true,
    "manager_visible": true
  },
  "explanation": {
    "status": "VERIFIED",
    "headline": "An electric vehicle may suit your travel pattern.",
    "customer_message": "Your recurring fuel spending and annual mileage suggest that an electric
vehicle may suit your travel pattern. The estimated potential reduction is 900-1,200 kg CO2e per year,
based on documented fuel and electricity factors. Explore EV options, subject to eligibility checks.",
    "manager_message": "This customer has high fuel spending, high annual mileage and an existing petrol
vehicle. EV Finance passed the configured eligibility rules and should be considered for the next
customer conversation.",
    "verification": {
```

```
      "passed": true,
      "verified_fields": [
        "product_name",
        "reason_codes",
        "impact_range",
        "impact_unit",
        "action_label",
        "data_cutoff_date"
      ],
      "unsupported_claims": [],
      "fallback_used": false
    },
    "llm": {
      "provider": "OPENAI",
      "model": "MODEL_NAME",
      "prompt_version": "customer_insight_v1.0.0"
    }
  },
  "audit": {
    "pipeline_status": "SUCCESS",
    "component_versions": {
      "feature_pipeline": "features_v1.0.0",
      "transaction_classifier": "transaction_classifier_v1.0.0",
      "propensity_model": "ev_propensity_v1.0.0",
      "eligibility_rules": "ev_eligibility_v1.0.0",
      "carbon_engine": "carbon_engine_v1.0.0",
      "llm_prompt": "customer_insight_v1.0.0",
      "llm_verifier": "verifier_v1.0.0"
    },
    "warnings": [],
    "errors": [],
    "human_review_required": false
  }
}
```

# 3. Fixed enum values

| Field | Allowed values |
|---|---|
| product_code | EV_FINANCE, GREEN_HOME_LOAN, SOLAR_FINANCE, PAPERLESS_BANKING |
| eligibility.status | ELIGIBLE, INELIGIBLE, INSUFFICIENT_DATA, MANUAL_REVIEW |
| recommendation.decision | RECOMMEND, DO_NOT_RECOMMEND, MANUAL_REVIEW |
| impact.status | AVAILABLE, INSUFFICIENT_DATA, NOT_APPLICABLE, CALCULATION_ERROR |
| explanation.status | VERIFIED, FALLBACK_USED, NOT_GENERATED, VERIFICATION_FAILED |
| confidence_band | LOW, MEDIUM, HIGH |
| priority_band | LOW, MEDIUM, HIGH |
| decision_codes[].result | PASS, FAIL, UNKNOWN |
| audit.pipeline_status | SUCCESS, PARTIAL_SUCCESS, FAILED |

## Optional products

GREEN_INVESTMENTS and GREEN_LOAN may be added later. Do not include them until the four required products work end to end.

## Code style

All machine-readable codes use uppercase SNAKE_CASE. Human-readable wording belongs in label, message or description fields.

# 4. Ineligible example

This customer has a strong model score, but already owns an EV and has no replacement use case. The recommendation is therefore blocked.


```
{
  "prediction": {
    "propensity": 0.91,
    "confidence": 0.89,
    "confidence_band": "HIGH",
    "model_version": "ev_propensity_v1.0.0",
    "reason_codes": []
  },
  "eligibility": {
    "status": "INELIGIBLE",
    "is_eligible": false,
    "rule_version": "ev_eligibility_v1.0.0",
    "decision_codes": [
      {
        "code": "EXISTING_EV_WITHOUT_REPLACEMENT_CASE",
        "result": "FAIL",
        "message": "Customer already owns an EV and no replacement case exists."
      }
    ],
    "failed_rules": [
      "EXISTING_EV_WITHOUT_REPLACEMENT_CASE"
    ],
    "missing_requirements": [],
    "requires_human_review": false
  },
  "impact": {
    "status": "NOT_APPLICABLE"
  },
  "recommendation": {
    "decision": "DO_NOT_RECOMMEND",
    "priority_score": null,
    "priority_band": null,
    "action_code": null,
    "action_label": null,
    "customer_visible": false,
    "manager_visible": true
  }
}
```

# 5. Missing-data example

When mileage is missing, eligibility cannot be finalized and the carbon engine cannot produce a trustworthy estimate. Null is used instead of an invented number.


```
{
  "eligibility": {
    "status": "MANUAL_REVIEW",
    "is_eligible": null,
    "rule_version": "ev_eligibility_v1.0.0",
    "decision_codes": [
      {
        "code": "ANNUAL_MILEAGE_MISSING",
        "result": "UNKNOWN",
        "message": "Annual mileage is not available."
      }
    ],
    "failed_rules": [],
    "missing_requirements": [
      "annual_mileage"
    ],
    "requires_human_review": true
  },
  "impact": {
    "status": "INSUFFICIENT_DATA",
    "method_id": "EV_AVOIDED_EMISSIONS_V1",
    "primary_metric": null,
    "supporting_metrics": [],
    "assumptions": [],
    "emission_factors": [],
    "calculation_inputs": {
      "annual_mileage": null,
      "annual_fuel_litres": null,
      "estimated_ev_kwh": null
    }
  },
  "recommendation": {
    "decision": "MANUAL_REVIEW",
    "priority_score": null,
    "priority_band": null,
    "action_code": "REQUEST_MILEAGE_DATA",
    "action_label": "Request mileage information",
    "customer_visible": false,
    "manager_visible": true
  }
}
```

NON-NEGOTIABLE

# 6. Mandatory implementation rules

1. propensity, confidence, importance, data_completeness and priority_score must be numbers from 0.0 to 1.0 when present.

2. Use null for unavailable values. Never use N/A, an empty string, zero or -1 to mean missing.

3. Use ISO 8601 dates and timestamps. Store the timezone whenever a time is included.

4. Use fixed uppercase SNAKE_CASE codes. Do not create spelling variants in different services.

5. Do not remove mandatory fields. Keep the field and set its value to null when the value is unavailable.

6. customer_visible may be true only when eligibility.status is ELIGIBLE and recommendation.decision is RECOMMEND.

7. An unavailable impact must not contain zero. Zero means a valid calculation produced zero; null means unknown.

8. Every available impact must include a unit, method, time horizon, assumptions and emissions-factor provenance.

9. The LLM must not modify customer, prediction, eligibility, impact or recommendation.

10. Every generated explanation must be verified. If verification fails, use a deterministic fallback template.

11. Every pipeline run must record component versions, warnings, errors and whether human review is required.

12. The official product and eligibility rules remain authoritative even when the propensity score is high.

**Validate this contract at every API boundary with JSON Schema or Pydantic.**

# 7. Final integration checklist

- Person 1 can create customer and opportunity fields without ML code.
- Person 2 writes prediction without adding eligibility logic.
- Person 3 writes eligibility and impact without calling an LLM.
- Person 4 generates explanation only from approved fields and verifies every factual claim.
- Person 5 can display eligible, ineligible and missing-data examples without special-case UI code.
- All services accept schema_version 1.0.0 and reject incompatible major versions.
- The same test object passes through local code, API responses, saved predictions and dashboard views.
- Automated tests cover valid, ineligible, missing-data and calculation-error records.

**Freeze this contract before parallel development begins.**

## Recommended repository location

```
contracts/opportunity_v1.schema.json
tests/fixtures/eligible_ev.json
tests/fixtures/ineligible_ev.json
tests/fixtures/missing_data_ev.json
```

The examples in this PDF should become automated test fixtures so every service works with exactly the same records.

FINAL TEAM CONTRACT

# Predict clearly.
# Decide safely.
# Calculate transparently.
# Explain only verified facts.

**Schema version 1.0.0**

Prepared for the Lloyds Megathon Glass-Box Green Insights solution.